



CCS Developer Challenge

1. Introduction

Welcome to the CCS Development Challenge.

The objective of this challenge is to evaluate your ability to design and implement a RESTful API using Python, as well as your ability to integrate it with a simple frontend application.

The primary focus of this assessment is backend development. The frontend is intended only to demonstrate that your API can be consumed correctly.

You may use Django REST Framework.

2. Backend Requirements

Build a REST API for a simple **User Management System**.

2.1 Authentication

Implement a login endpoint using the following credentials:

```
[
  {
    "username": "user123",
    "password": "Password1!"
  },
  {
    "username": "admin",
    "password": "Admin123!"
  }
]
```

Requirements:

1. Authenticate users through the backend.
2. JWT authentication is preferred but not required.
3. Protect all User Management endpoints after authentication.

2.2 Database

Use one of the following databases:

1. PostgreSQL.

Requirements:

1. Use an ORM.
2. Include database migrations if supported by your framework.

3. Populate the database with at least **20 sample users** using a seed script, fixture, or migration.

Each user should contain the following fields:

id
first_name
last_name
email
phone
country
birth_date

2.3 REST API

Implement the following endpoints:

POST /login

GET /users

GET /users/{id}

POST /users

PUT /users/{id}

DELETE /users/{id}

The frontend must consume these endpoints exclusively.

2.4 Validation

Implement validation for:

1. Required fields.
2. Valid email format.
3. Duplicate email addresses.
4. Invalid user IDs.

2.5 Error Handling

Return appropriate HTTP status codes.

Examples:

- 200 OK
- 201 Created
- 400 Bad Request
- 401 Unauthorized
- 404 Not Found
- 500 Internal Server Error

Return meaningful error messages when applicable.

2.6 Project Structure

Organize the project following the conventions of the selected framework.

Separate responsibilities appropriately, for example:

1. Models.
2. Routes or Controllers.
3. Services or Business Logic.
4. Database configuration.
5. Validation.

3. Frontend Requirements

The frontend should be a lightweight client whose primary purpose is to consume the backend API.

Angular is preferred, but React or Vue are also accepted.

3.1 Design Reference

Use the following Figma as a visual reference:

<https://www.figma.com/design/7On7sfJUEmtwefH2db6yG/FRONTEND?node-id=112-2&p=f&t=Lhc6ZkXsVCsF1ZGb-0>

The objective is **not** to reproduce the design pixel by pixel. Functionality and API integration will be evaluated over visual accuracy.

Only the following screens are required:

1. Login.
2. People List.
3. User Details.

The Dashboard and Find Person screens are provided only as design references and are **not required**.

3.2 Login

Requirements:

1. Username field.
2. Password field.
3. Authenticate against the backend API.
4. Redirect to the People List after a successful login.

3.3 People List

Display a table containing at least the following fields:

1. Full Name.
2. Email.
3. Country.
4. Phone.

Include the following actions:

1. View user details.
2. Edit user.
3. Delete user.

The list should retrieve its data from the backend API.

Basic pagination or search is optional.

3.4 User Details

Display all available information for the selected user.

The page should allow editing the user's information and saving the changes through the backend API.

4. Minimum Frontend Requirements

If Angular is used, demonstrate the following concepts:

1. Standalone Components.
2. Services.
3. HttpClient.
4. Routing.
5. Reactive Forms.

The following features are optional and will not be evaluated:

1. Angular Signals.
2. Interceptors.
3. Angular Material.
4. Tailwind CSS.
5. State management libraries.

5. Bonus Features (Optional)

Candidates may implement any of the following:

1. Pagination.
2. Search or filtering.
3. Docker.
4. Unit tests.
5. Swagger / OpenAPI documentation.
6. Refresh Tokens.
7. Role-based authorization.
8. Logging.
9. Environment configuration.

6. Evaluation Criteria

Backend (70%)

Candidates will be evaluated on:

1. API design.
2. Authentication.
3. Database design.
4. CRUD implementation.
5. Validation.
6. Error handling.
7. Code quality.
8. Project organization.
9. Maintainability.

Frontend (30%)

Candidates will be evaluated on:

1. API integration.
2. Routing.
3. Forms.
4. Component organization.
5. Basic user experience.

7. Testing

Testing is optional.

Candidates may include unit or integration tests if desired.

8. Submission Guidelines

Please submit your completed challenge as follows:

1. Host your project in a public Git repository (GitHub, GitLab, or Bitbucket).
2. Include a **README.md** containing:
 1. Project overview.
 2. Prerequisites.
 3. Backend installation instructions.
 4. Frontend installation instructions.
 5. Database migration instructions.
 6. How to populate the database with sample users.
 7. Any assumptions or design decisions.
 8. Optional features implemented.
3. Include all migration and seed files required to run the project.
4. (Optional) Deploy the application and include the URL in the README.

9. General Notes

1. Clean, readable, and maintainable code is preferred over implementing unnecessary features.
2. The backend carries more weight in the evaluation than the frontend.
3. Follow the best practices of the framework you choose.
4. The application should be easy to run by following the instructions provided in the README.
5. The completed challenge should demonstrate your ability to build, document, and expose a production-style REST API.

10. Attachments



CHARLIE'S CAT LOVERS CLUB

Welcome Back

Enter your username and password to sign in

Username

Your username

Password

Your password

☐ Show password

SIGN IN

Having trouble signing in? [Contact Support](#)



Dominique Ch. ▾



Dashboard

People



Total People
75

5 than last week



Administration

Total Employee: 8

Updated 2 days ago



CEO

Total Employee: 1

Updated 2 days ago



Sales

Total Employee: 245

Updated 2 days ago



Marketing

Total Employee: 11

Updated 2 days ago



IT

Total Employee: 3

Updated 2 days ago



Finance

Total Employee: 8

Updated 2 days ago



Legal

Total Employee: 3

Updated 2 days ago



PPS

Total Employee: 11

Updated 2 days ago

List of active people

Delivery billable	Delivery unbillable	Administration	CEO	Sales	Marketing	Finance	HR	IT	Legal	PPS	L & D	OG	O&S	EBO
233	11	8	1	15	11	8	4	3	2	15	1	5	2	0

Delivery: 581 Non-delivery: 132 Total: 713

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COMPUTER
SYSTEMS**
Version: 1.0.0.11

Dominique Ch.

Dashboard

People

Find person

All people

List of active people

Export to Excel

Total People	Administration	CEO	Sales	Marketing	Finance	HR	IT	Legal	PPS	L & D	OG	OBS	EBO
	8	1	15	11	8	4	3	2	15	1	5	2	0

Delivery: 581 Non-delivery: 132 Total: 713

Filter

Regions

☒ UK (111)

☐ US (98)

☐ France (45)

Branches

☒ London (111)

☒ Liverpool (12)

☒ Manchester (5)

☒ Leeds (3)

☐ Cardiff (1)

Divisions

☒ Management (43)

☒ Finance (10)

☒ Legal (1)

☒ Operation (1)

☐ IT (1)

Clear

Found: 110

Full name	Branch	Division	User type	Contract type	Start date	ID
Konstantin Konstantinopolsky	London	Management	Delivery billable	Employee	12.01.2013	14321
Andrew Salgado	London	Management	Delivery billable	Employee	21.04.2014	21345
Magdalena Edinborough	Liverpool	Operation	Delivery billable	Subcontractor	02.11.2015	1894
Matt Travis	Liverpool	Management	Delivery billable	Employee	02.11.2016	135567
Daniel Wellington	Manchester	Operation	Delivery billable	Employee	02.11.2012	10973
Lucia Mirocini	Leeds	Finance	Delivery billable	Employee	02.11.2011	6324
Anette Brown	London	Operation	Delivery billable	Employee	02.11.2005	4322
Christian Lambrou	London	Finance	Delivery billable	Employee	02.11.2009	4521
Phil Collins	Manchester	Finance	Delivery billable	Subcontractor	02.11.2008	1422
Winston Holliday	London	Operation	Delivery billable	Employee	02.11.2017	45667
Carla Andrade	London	Operation	Delivery billable	Employee	02.11.2017	98871

Previous

12345...31Next

Show all

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Dominique Ch.

Dashboard

People

Find person

All people

Filter

Name

Andrew Salgado

Email address

example@mail.com

Login

Login

Branch

Select

1C Id

Id

Practice

Select

User type

Any

User contract type

Any

CV

Search...

☒ Capability

☒ Technology

☒ Certificate

☒ Course

Capabilities

Search...

ANDOR

Technology

Search...

ANDOR

Certificates

Search...

ANDOR

Courses

Search...

ANDOR

Education

Select

Search for specialization

☒ Show User Type, Login, Start Date, 1C Id

☐ Show current assignments

☐ Show Skype, Office phone, Mobile phone, Room number, Title

☐ Include terminated employees

ClearSearch

COURTHOUSE
COMPUTER
SYSTEMS

Version: 1.0.0.11



Dominique Ch. ▾





Dashboard



People

Details 

Name

Dominique Charpentier

Title

CEO

Practice

Finance

Branch

Paris ▾

Contract

Employee

Grade

MG3

Division

Moonlight - Paris

Division manager

Login

id/ceo

Status history

Activated

13/05/2009



Dominique Charpentier

CEO



Calendar

Vacations

Timesheet

Corporate CV



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Version: 1.0.0.11



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